UNIT QUESTIONS

SKILLS CRITICAL THINKING 5

1 These three organelles are found in cells: nucleus, chloroplast and mitochondrion.

a Which of the above organelles would be found in:

i a cell from a human muscle? **(1)**

ii a palisade cell from a leaf? **(1)**

iii a cell from the root of a plant? **(1)**

SKILLS REASONING

b Explain fully why the answers to ii) and iii) above are different. **(1)**

SKILLS INTERPRETATION

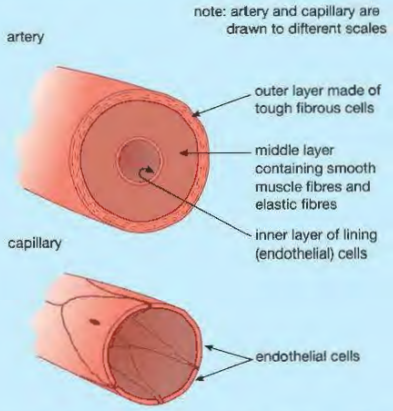
c What is the function of each organelle? **(3)**

Total 7 marks

SKILLS INTERPRETATION, REASONING 5

2 In multicellular organisms, cells are organised into tissues, organs and organ systems.

a The diagram shows a section through an artery and a capillary.



Explain why an artery can be considered to be an organ whereas a capillary cannot. **(2)**

b Organ systems contain two or more organs whose functions are linked. The digestive system is one human organ system. (See Chapter 4.)

SKILLS CRITICAL THINKING 4

i What does the digestive system do? **(2)**

5

ii Name three organs in the human digestive system. Explain what each organ does as part of the digestive system. **(6)**

4

iii Name two other human organ systems and, for each system, name two organs that are part of the system. **(6)**

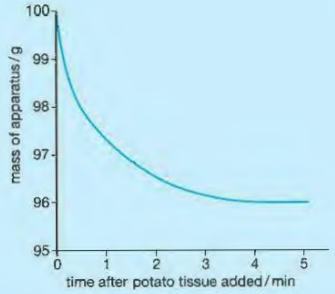
Total 16 marks

SKILLS ANALYSIS

3 Catalase is an enzyme found in many plant and animal cells. It catalyses the breakdown of hydrogen peroxide into water and oxygen.

$$\text{hydrogen peroxide} \xrightarrow{\text{catalase}} \text{water} + \text{oxygen}$$

a In an investigation into the action of catalase in potato, 20 g of potato tissue was put into a small beaker containing hydrogen peroxide weighing 80 g in total. The temperature was maintained at 20 °C throughout the investigation. As soon as the potato was added, the mass of the beaker and its contents was recorded until there was no further change in mass. The results are shown in the graph.



5

i How much oxygen was formed in this investigation? Explain your answer. **(2)**

6

ii Estimate the time by which half this mass of oxygen had been formed. **(2)**

SKILLS REASONING 8

iii Explain, in terms of collisions between enzyme and substrate molecules, why the rate of reaction changes during the course of the investigation. **(2)**

b The students repeated the investigation at 30 °C. What difference, if any, would you expect in:

SKILLS DECISION MAKING 9

i the mass of oxygen formed?

ii the time taken to form this mass of oxygen?

Explain your answers. **(4)**

Total 10 marks

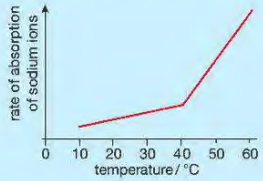
SKILLS CRITICAL THINKING 7

4 Different particles move across cell membranes using different processes.

a The table below shows some ways in which active transport, osmosis and diffusion are similar and some ways in which they are different. Copy and complete the table with ticks and crosses. **(3)**

Feature	Active transport	Osmosis	Diffusion
particles must have kinetic energy			
requires energy from respiration			
particles move down a concentration gradient			

b The graph shows the results of an investigation into the rate of diffusion of sodium ions across the membranes of potato cells.



SKILLS ANALYSIS, REASONING (6)

i Explain the increase in the rate of diffusion up to 40 °C. **(2)**

ii Suggest why the rate of increase is much steeper at temperatures above 40 °C. **(2)**

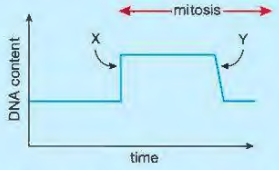
Total 7 marks

5

SKILLS REASONING (8)

Cells in the wall of the small intestine divide by mitosis to replace cells lost as food passes through.

a Chromosomes contain DNA. The graph shows the changes in the DNA content of a cell in the wall of the small intestine as it divides by mitosis.

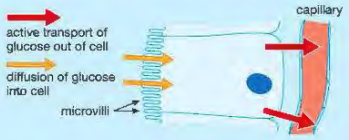


i Why is it essential that the DNA content is doubled (X) before mitosis begins? **(2)**

SKILLS ANALYSIS (8)

ii What do you think happens to the cell at point Y? **(1)**

b The diagram shows a cell in the wall of a villus in the small intestine. Some of the processes involved in the absorption of glucose are also shown.



SKILLS INTERPRETATION (6)

i What is the importance of the small intestine having villi? **(1)**

(7)

ii Suggest how the microvilli adapt this cell to its function of absorbing glucose. **(1)**

(8)

iii Suggest how the active transport of glucose out of the cell and into the blood stream helps with the absorption of glucose from the small intestine. **(2)**

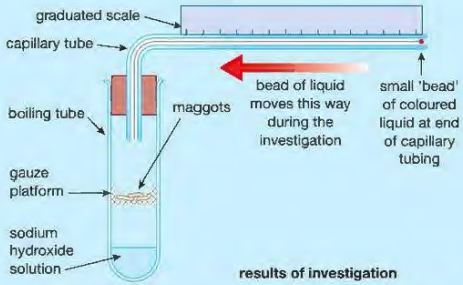
Total 7 marks

6

SKILLS REASONING (8)

A respirometer is used to measure the rate of respiration. The diagram shows a simple respirometer. The sodium hydroxide solution in the apparatus absorbs carbon dioxide. Some results from the investigation are also shown.

> **!** Safety Note: Eye protection should be worn when setting up the apparatus as sodium hydroxide is very hazardous to the eyes.



results of investigation

Experiment	Distance moved by bead / mm
1	20
2	3
3	18

a Assume that the maggots in the apparatus respire aerobically.

i Write the symbol equation for aerobic respiration. **(4)**

ii From the equation, what can you assume about the amount of oxygen taken in and carbon dioxide given off by the maggots? Explain your answer. **(3)**

iii Result 2 is significantly different from the other two results. Suggest a reason for this. **(2)**

(9)

iv How would the results be different if the organisms under investigation respired anaerobically? **(2)**

Total 11 marks

7

SKILLS INTERPRETATION (5)

The table below shows some features of different groups of organisms. Copy and complete the table by putting a tick in the box if the organism has that feature, or a cross if it lacks the feature.

Feature	Type of organism		
	Plant	Fungus	Virus
they are all parasites			
they are made up of a mycelium of hyphae			
they can only reproduce inside living cells			
they feed by extracellular digestion by enzymes			
they store carbohydrate as starch			

Total 5 marks

SKILLS CRITICAL THINKING 6

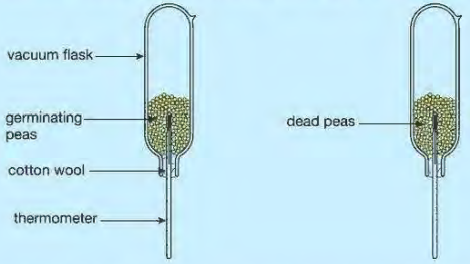
8 Copy and complete the following account.

Plants have cell walls made of _____ . They store carbohydrate as the insoluble compound called _____ or sometimes as the sugar _____ . Plants make these substances as a result of the process called _____ . Animals, on the other hand, store carbohydrate as the compound _____ . Both animals' and plants' cells have nuclei, but the cells of bacteria lack a true nucleus, having their DNA in a circular chromosome. They sometimes also contain small rings of DNA called _____ , which are used in genetic engineering. Bacteria and fungi break down organic matter in the soil. They are known as _____ . Some bacteria are pathogens, which means that they _____ .

Total 8 marks

SKILLS DECISION MAKING 6

9 The diagram shows the apparatus used to investigate germination of pea seeds. At the start of the experiment the temperature in both flasks was 19 °C



The apparatus was left for 24 hours. At the end of this time the temperature in the flask with germinating peas was 22 °C, while the temperature in the flask with dead peas was 19 °C.

a Explain the biological reason for this difference in temperature. **(2)**

b The seeds in both flasks were washed in disinfectant before the experiment. Explain why this was done. **(1)**

c Cotton wool was used to hold the seeds in the flasks. Suggest why cotton wool was used instead of a rubber bung. **(1)**

SKILLS EXECUTIVE FUNCTION 5

d Pea seeds were used in both flasks. State another variable that should have been controlled in the experiment. **(1)**

Total 5 marks

SKILLS EXECUTIVE FUNCTION 7

10 A piece of meat is a tissue composed of muscle fibres. Muscle fibres use ATP when they contract. Describe an investigation to find out if a solution of ATP will cause the contraction of muscle fibres.

Total 6 marks